Main Wheels (Mod 6/2002) – General Data

1. General

The wheel is cast magnesium and conforms to all tire and rim association standards for a 11.00-12 divided type wheel, suitable for use with all 11.00-12, 10 ply tires. It is a tubeless design only. The wheel incorporates inboard and outboard halves which are fastened together with bolts, washers, and nuts. The brake disc is attached to the wheel by bolts. The wheel rotates on two tapered roller bearings, which seat in bearing cups in the wheel half hubs. Rubber lip seals provide protection and lubricant retention for the bearing.

2. Removal/Installation

A. Remove Main Wheel

1) Remove the brake assembly and place it on a clean flat surface, as per 32-40-31, Wheel Brakes Unit (Mod 6/2002) – General Data.

2) Jack aircraft until wheel is clear of ground. (Refer to 07-00-00, Lifting and Shoring – General Data).

3) Deflate tire.

4) Remove snap ring and hubcap.

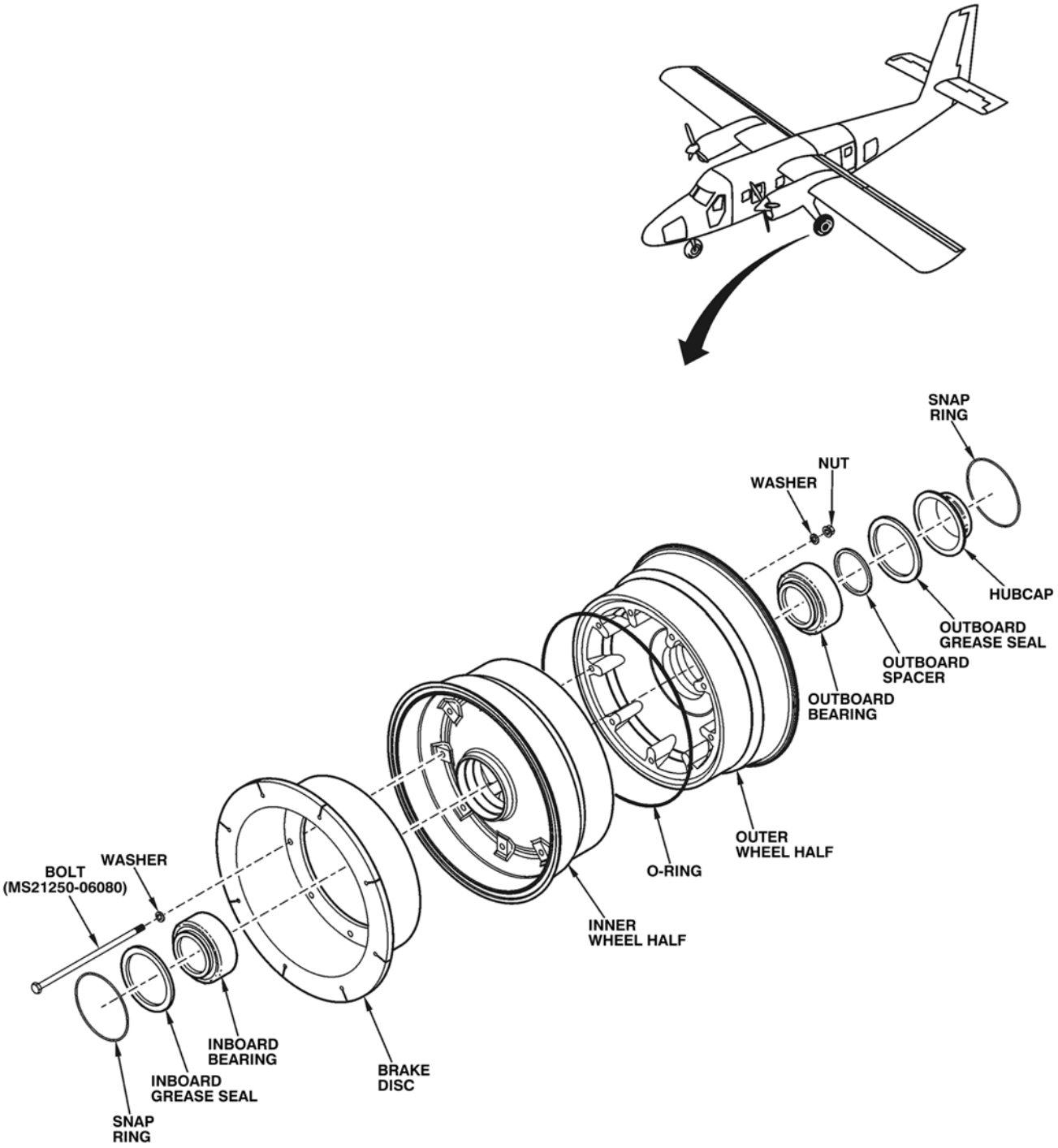
5) Remove two safety bolts and self locking nuts from axle nut. Remove axle nut using wrench SD5523.

6) Remove outboard grease seal, spacer and bearing.

Note Place wheel seals, spacer and bearings on a clean, flat surface to prevent any contamination.

7) Carefully slide wheel/tire assembly off of axle.

8) Remove inboard bearing, seal and retaining clip.



Main Wheel Assembly
Figure 1

B. Install Main Wheel

1) Apply a thin coat of grease (MIL–PRF–81322) or MOBIL SHC 100 (refer to 12-20-10, Lubrication – General Data, Table 1) to axle and axle threads.

2) Install lubricated inboard wheel bearing in wheel assembly with grease seal and retaining clip.

3) Carefully slide wheel/tire assembly on to axle.

4) Install lubricated outboard bearing into wheel assembly with spacer and grease seal.

a) Install axle nut using wrench SD5523 and torque to 150 in–lbs to 200 in–lbs while seating the bearing.

b) Back off axle nut.

c) Retighten to 30 in–lbs to 40 in–lbs while rotating.

d) Rotate axle nut to nearest axle slot and install two safety bolts and self locking nuts.

Note Wheel must rotate freely, without perceptible play.

5) Install hubcap and snap ring.

6) Inflate tire.

7) Refer to 32-40-31, Wheel Brakes Unit (Mod 6/2002) – General Data, to reinstall brake assembly.

Note If installation of main wheel on aircraft is part of gear configuration change then do step as per Para 2.B.8).

8) Do procedure for gear configuration change as per 32-10-11, Main Gear Leg – General Data, Para 2.C.

C. Remove Main Wheel Tire (Tubeless)

1) Remove wheel from aircraft. (Refer to Para 2.A).

2) Release all pressure from tire by slackening air valve core then finally remove core completely.

3) Using tire demounters, wooden wedges, or an approved tire machine, separate edges of tire from wheel hub beading.

4) Remove bolts, self-locking nuts, washers and countersunk washers, holding halves of hub together. (Refer to Figure 1).

5) Carefully separate hub halves, and remove O-ring and tire. Check O-ring for serviceability and discard if damaged.

D. Install Main Wheel Tire (Tubeless)

1) Check hub for scratches, scores and cleanliness according to OVH Manual 040–21101. Clean tire area of hub with a dry cleaning solvent. Dust area with french chalk (talc).

2) Assemble wheel on a clean flat surface according to OVH Manual 040–21101.

3) Position hub half containing air valve on bench. Lightly lubricate serviceable O-ring with silicone grease (DOW Molykote 55M) and install on hub. Position serviceable tire on hub with red spot on tire aligned with air valve.

CAUTION ENSURE BOLTS ARE MS21250–06080. DO NOT USE BOLTS, VALVE STEM BUSHING AND BRAKE DISC SPACER FROM S.O.O. 6048 AND MOD 6/2183.

4) Lubricate threads of bolts and bearing surfaces of nuts, bolt heads and washers with anti-seize compound (MIL–T–5544A).

WARNING **DO NOT INFLATE TIRE UNTIL ALL BOLTS HAVE BEEN PROPERLY TIGHTENED. IT IS RECOMMENDED THAT ALL MOUNTED TIRES BE INFLATED INSIDE A METAL CAGE TO AVOID POSSIBLE INJURY TO PERSONNEL.**

CAUTION **DO NOT USE IMPACT OR POWER WRENCHES TO TIGHTEN OR TORQUE WHEEL BOLTS OR NUTS.**

5) Install core in air valve and inflate tire to correct pressure. (Refer to 12-10-20, Landing Gear – General Data).

6) Apply a 1” x 3” creep mark using a suitable contrasting coloured paint between tire and rim.

7) Install wheel on aircraft. (Refer to Para 2.B).

3. Inspection/Check

A. Inspect Main Wheel Hub

1) Remove main wheel and main wheel tire. (Refer to Para 2.A and Para 2.C).

2) Clean hub halves with dry cleaning solvent (P–D–680 or equivalent) and dry with filtered and dried compressed air. Refer to OVH Manual 040–21101.

3) Inspect hub halves for cracks (using the fluorescent penetrant method or an approved dye penetrant method), chipping of rims, damage and corrosion. Replace hub if cracks or chipping of rims is evident. Refer to OVH Manual 040–21101.

Note Small nicks or gouges can be blended out and polished with fine aluminum oxide cloth (refer to Manufacturer's Overhaul Manual for definition and limitation of damage). Refer to OVH Manual 040–21101.

4) Inspect bearings and replace if damaged. Refer to OVH Manual 040–21101.

5) Inspect brake disc wheel bolts and nuts as per OVH 030–17000.

6) Install tire to main wheel and install main wheel on aircraft. (Refer to Para 2.B and Para 2.D).